Pampanga State Agricultural University
Magalang, Pampanga

PAMPANGA STATE AGRICULTURAL UNIVERSITY
MAGALANG PAMPANGA
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August 01, 2024

DR. ANITA G. DAVID
SUC President IV
PSAU, Magalang, Pampanga

Dear Dr. David:

A blissful day!

Given the implementation of the PhilRice-funded research project entitled **"Product Development and Evaluation of Balakat [*Ziziphus talanai* (Blanco) Merrill] Nano-Molluscicide Against Invasive Apple Snails [*Pomacea canaliculata* (Lamarck) in Direct-Seeded Rice,"** may I have the honor to recommend **for the approval of the service of the following Professional Expert**.

As such, she is recommended to receive a professional fee at the rate opposite name, chargeable against the project budget, Fund 07 (PhilRice Balakat-Nano). - Professional Expert/Services Fees allotment.

Name	Assignment	Duration	Rate/Consultation	No. of consultations
Christine Joy Ann Padiernos	Chemist	August 01, 2024, to February 28, 2025	3,000.00	3

Please find the attached work plan and the MOA for your reference.

Thank you very much.

Very truly yours,

EVELYN V. TOTAAN, Ph.D.
Project Leader

Noted:

ROGELIO P. CARANDANG, JR., DVM, Ph.D.
Director, Research and Development Office

Recommending Approval:

GERALDINE C. SANCHEZ, Ph.D.
Vice President for RIET

SYLVIA S. DE GUZMAN, Ed.D.
Vice President for Academic Affairs

MINA LYN G. ALVIZ. M.A.Ed.
Acting Vice President for Administration and Finance
Fund 07 Philrice Balakat fpla

APPROVED:

ANITA G. DAVID, Ed.D.
SUC President IV

Republic of the Philippines
PAMPANGA STATE AGRICULTURAL UNIVERSITY
PAC, Magalang, Pampanga

PROFESSIONAL EXPERT'S WORKPLAN

Name: CHRISTINE JOY ANN PADIERNOS
Office/Unit/Project: Fund 07 (PHILRICE NANO BALAKAT)

ACTUAL DUTIES/ WORK BEING DONE DAILY	NO. OF HOURS SPENT DOING THE TASK/S	TARGET OUTPUT	ACTUAL DUTIES/ WORK BEING DONE EVERY WEEK	NO. OF HOURS SPENT DOING THE TASK/S	TARGET OUTPUT	ACTUAL DUTIES/ WORK BEING DONE EVERY MONTH	NO. OF HOURS SPENT DOING THE TASK/S	TARGET OUTPUT
			Assist in conducting experiments (Biosynthesis of Silica NPs infused with aqueous extracts of *Z. talanai*, Optimization Studies, and Molluscicidal evaluation), Pellet formulation development, small-scale and large-scale field experiments	6 hrs	Assisted the Project Staff in the conduct of the different tests/ experiments/ assays, small-scale and large-scale field experiments	Prepare accomplish report	2 hrs	Prepared and submitted Monthly Accomplish Report
			Other project-related tasks may be assigned from time to time.	2 hrs				
Total No. of hours:			**Total No. of hours: 8 hours**			**Tota No. of hours:**	**2 hours**	

Prepared by:

EVELYN V. TOTAAN, Ph.D.
Name and Signature of the Project Leader

Conforme:

CHRISTINE JOY ANN PADIERNOS
Name and Signature of the Staff

Recommending Approval:

GERALDINE C. SANCHEZ, Ph.D.
Vice President for RIET

MINA LYN G. ALVIZ, M.A.Ed.
Acting Vice President for Administration and Finance

Approved by:

ANITA G. DAVID, Ed.D.
President SUC IV

Control No.
MOA-2023-03-114

MEMORANDUM OF AGREEMENT

KNOW ALL MEN BY THESE PRESENTS:

This **Memorandum of Agreement (Agreement)** is entered into by and between:

The **PHILIPPINE RICE RESEARCH INSTITUTE**, a government corporate entity under the Department of Agriculture, with office address at Maligaya, Science City of Muñoz, Nueva Ecija, represented herein by its Executive Director, **DR. JOHN C. DE LEON**, hereinafter referred to as **"DA-PhilRice"**;

JOHN C. DE LEON

ANITA G. DAVID

and

The **PAMPANGA STATE AGRICULTURAL UNIVERSITY,** a government institution of higher learning in education, research and extension, with office address at Magalang, Pampanga, represented herein by its President, **DR. ANITA G. DAVID,** hereinafter referred to as **"PSAU"**.

(**DA-PhilRice** and **PSAU** are also individually referred to as "**Party**" and collectively as "**Parties**".)

EDUARDO JIMMY P. QUILANG

GERALDINE C. SANCHEZ

WITNESSETH:

WHEREAS, DA-PhilRice, as the national rice research for development (R4D) agency, aims to improve farm productivity, profitability, and sustainability, and is tasked to generate rice and rice-based farming technologies;

WHEREAS, **PSAU** is an institution that provides quality science education and professional services through collaborative research and development, equipped with highly competent staff and state-of-the-art accredited facilities;

WHEREAS, **PSAU** implements the research study **Product Development and Evaluation of Balakat Nano-Molluscicide Against *Pomacea canaliculata* in Direct-Seeded Rice**; and

WHEREAS, DA-PhilRice is willing to collaborate with **PSAU** in the implementation of the above-mentioned research study.

NOW, THEREFORE, for and in consideration of the aforementioned premises, the Parties have agreed as follows:

ARTICLE 1
AREAS OF COLLABORATION

JENNIFER T. NIONES

EVELYN V. TOTAAN

The collaboration will focus on the implementation of "**Product Development and Evaluation of Balakat Nano-Molluscicide Against *Pomacea canaliculata* in Direct-Seeded Rice.**" The research study will be implemented from **March 01, 2023** to **February 28, 2025,** following the work plan schedule (Annex 2 – **Workplan Schedule implementation**).

ARTICLE 2
FUNDING, FINANCIAL RECORDS, AND REPORTS

Section 1. *Provision of funds* – **DA-PhilRice** shall provide funds amounting to **Two million six hundred thousand pesos (PhP 2,600,000.00)** for the implementation of the research study (details of budget shown in Annex 1 – **Budget schedule**). The fund will be transferred based on the scheme below:

Table 1. Schedule of fund release to **PSAU**

Fund Release	Year	Amount	Schedule
First tranche	1	Php 1,200,000.00	Upon signing of the MOA by both parties
Second tranche	2	Php 1,400,000.00	Upon submission of accomplishment and financial reports duly received by COA for the first tranche

Section 2. *Compliance with standards of government rules and regulations* – All items for financing shall be subjected to existing government accounting and auditing law, rules and regulations.

Section 3. *Management of activity funds* – **PSAU** shall manage and be accountable for the use of funds released to it. **PSAU** shall use the funds solely for the purpose of implementing the activities of the research study specified in the workplan (Annex 2).

Section 4. *Financial records* – **PSAU** shall keep and maintain financial records, books of accounts, and supporting receipts and documents for the funds provided under this **Agreement** in accordance with standard accounting and auditing rules and regulations.

Section 5. *Right of inspection* – **DA-PhilRice** shall have the right to inspect the financial records and books of accounts, together with the supporting receipts and documents pertaining to the disbursement of funds provided for under this **Agreement**.

Section 6. *Liquidation and financial reports* – **PSAU** shall submit the final audited financial report to **DA-PhilRice** covering all fund releases upon completion of the research study (**March 31, 2025**). A financial report shall be duly verified by the **PSAU** Resident Auditor, subject to the final approval of **DA-PhilRice**.

Section 7. Equipment – All equipment purchased shall be the property of **DA-PhilRice,** which, however, may be donated to **PSAU** after the implementation of the research study.

Section 8. *Accomplishment Reports* – **PSAU** shall submit progress/accomplishment reports according to the schedule (Table 2). **DA-PhilRice** and **PSAU** shall prepare and submit a scientific or technical paper(s) based on the research study's results for publication in a refereed journal.

JOHN C. DE LEON

ANITA G. DAVID

EDUARDO JIMMY P. QUILANG

GERALDINE C. SANCHEZ

JENNIFER T. NIONES

EVELYN V. TOTAAN

Table 2. Schedule of submission of reports

Activities	Date of Submission
Submission of Study Progress Report and R & D Highlights (Matrix and Narrative Forms)	Within 15 days after the end of each quarter
Submission of Terminal Report, Scientific/ technical paper, Patent Application to PhilRice	Within 30 days after the end of the project

JOHN C. DE LEON

ANITA G. DAVID

ARTICLE 3
MANAGEMENT, IMPLEMENTATION, MONITORING, AND EVALUATION OF ACTIVITY

The research study shall be implemented according to the protocol and the Parties hereto agree to do and perform as follows:

1. **DA-PhilRice** shall:
   1.1 Provide funds for the conduct of the research study in the amount **of TWO MILLION SIX HUNDRED THOUSAND PESOS (PhP 2,600,000.00)**;
   1.2 Conduct a review of the progress of the research study for proper action; and
   1.3 Monitor and evaluate the activities according to protocol and targets indicated in the Research Study Design Matrix (Annex 2).

2. **PSAU** shall:
   2.1 Issue an official receipt to **DA-PhilRice** for the fund release;
   2.2 Administer and disburse the funds as specified in Item 1 hereof in accordance with the work plan schedule, expenditure items stipulated in Annex 2 following government accounting law, rules, and regulations;
   2.3 Designate the project leader and staff who are regular staff to receive corresponding incentive, and hired personnel with corresponding salary only as stipulated in Annex 2;
   2.4 Submit audited financial report received and stamped by the COA Resident Auditor;
   2.5 Return to **DA-PhilRice** any unutilized funds, which must be reflected in the financial statements; and
   2.6 Submit quarterly and terminal reports and **manuscript(s) of paper(s) for publication/Patent Application** to **DA-PhilRice** together with all raw and processed data and information collected for the research study as indicated in Article 2, Section 8.

EDUARDO JIMMY P. QUILANG

GERALDINE C. SANCHEZ

ARTICLE 4
INTELLECTUAL PROPERTY RIGHTS

Section 1. *Ownership*. Intellectual property rights, their derivatives, and other proprietary information generated by the activity shall be jointly owned by **PSAU** and **DA-PhilRice**.

JENNIFER T. NIONES

EVELYN V. TOTAAN

Section 2. *Recognition of principal authors and institutions.* – All reports or articles, including tri-media presentations using activity data and output, shall recognize the principal researchers and authors involved in the activity and acknowledge **DA-PhilRice** as the primary agency involved in supporting and implementing the activity.

Section 3. *Management of Intellectual Property Rights (IPRs) and Authority to Enter into Licensing Agreements.* While IPRs generated are jointly owned by **PSAU** and **DA-PhilRice**, the latter is authorized and assigned the rights to manage, register, deposit, and with the concurrence of **PSAU**, negotiate, execute, and enter into technology transfer arrangements, and collect and disburse royalties from such arrangements.

ARTICLE 5
FORCE MAJEURE

No party shall be liable to the other for failure to perform its obligations hereunder due to the occurrence of any event beyond the reasonable control of such party including but not limited to government regulations and orders, acts of God, riots, strikes, civil commotions, and any other similar cause(s).

ARTICLE 6
AMENDMENTS

No modifications of the **Agreement** or any of its provisions shall be valid except by a written agreement of the contracting parties.

ARTICLE 7
EFFECTIVITY AND FINAL PROVISIONS

This **Agreement** shall take effect immediately upon signing and will remain effective for a period of **24 months**, unless extended or revoked in writing upon the consent of both parties. Either party shall provide notice to the other party 30 days before its termination date should it decide to prematurely terminate this **Agreement**.

IN WITNESS WHEREOF, the Parties hereunto have affixed their signatures on the dates and place indicated below:

Department of Agriculture - Philippine Rice Research Institute	**Pampanga State Agricultural University**
JOHN C. DE LEON *Executive Director*	**ANITA G. DAVID** *SUC President IV*
Place: _____	Place: _____
Date: _____	Date: _____

PHILRICE Philippine Rice Research Institute
CPD23-0281

Signed in the presence of:

EDUARDO JIMMY P. QUILANG
OIC, Office of the Deputy Executive Director for Research

GERALDINE C. SANCHEZ
Vice President for Research, Extension & Training

JENNIFER T. NIONES
Head, Crop Protection Division

EVELYN V. TOTAAN
Professor VI, College of Arts and Sciences, Project Leader

CERTIFICATION OF FUNDS AVAILABILITY

Certified funds available amounting to **PhP 1,200,000.00** for Year 1 implementation.

MARY GRACE DG. CORPUZ
Head, Financial Management Division

ACKNOWLEDGMENT

Republic of the Philippines)
Province of Nueva Ecija) S.S.
Science City of Muñoz)

BEFORE ME, a Notary Public for and in above jurisdiction, personally appeared the following:

Name	ID No.	Date Issued	Place Issued
ANITA G. DAVID	P2700193B	02 Aug 2019	DFA Pampanga
JOHN C. DE LEON	90-0501/May/1990/SCMNE		PhilRice ID

known to me to be the same persons who executed the foregoing instrument consisting six (6) pages including this page where the acknowledgment is written. The parties and their instrumental witnesses signed all the pages thereof and acknowledged to me that the same is their own free and voluntary act and deed, and that of the entities which they respectively represent.

IN WITNESS WHEREOF, I have hereunto set my hand and seal this ___ day of MAR 2 2 2023 _____, 2023 at _____, Philippines.

RODOLFO A. TOBIAS
Notary Public
Until December 31, 2024
Notarial Commission No. SD2023-002
PTR No. 6875148; 01-03-2023
Science City of Muñoz, Nueva Ecija
Roll No. 33624
MCLE Compliance No. VIII-0001315

Doc. No. 113;
Page No. 23;
Book No. 379;
Series of 2023.

ANNEX 1. BUDGET

Maintenance and Other Operating Expenses	**Total**
Year 1	
A. Direct Cost	
1. Traveling expenses	11,360.00
2. Communication expenses	12,000.00
3. Supplies and materials	300,000.00
4. MOOE	
4.1 Service contracts (Research Assistant)	258,840.00
4.2 Honoraria (Researchers)	228,000.00
4.3 Others	389,800.00
SUB-TOTAL for MOOE	**PhP 1,200,000.00**
Year 2	
A. Direct Cost	
1. Traveling expenses	18,000.00
2. Communication expenses	16,000.00
3. Supplies and materials	385,160.00
4. MOOE	
4.1 Service contracts (Research Assistant)	258,840.00
4.2 Honoraria (Researchers)	228,000.00
4.3 Others	494,000.00
SUB-TOTAL for MOOE	
TOTAL	**PhP 1,400,000.00**
GRAND TOTAL	**PhP 2,600,000.00**

ANNEX 2. WORKPLAN SCHEDULE

Study	Activity	2023				2024			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
	Preparatory Activities								
	Inception Meeting								
	Team Organization								
	Finalization of work schedule and Workplan								
	MOA/MOU Signing								
	Administrative and Financial Works								
1	**Biosynthesis of Silica NPs infuse with aqueous extracts of *Z. talanai indica***								
	Collection of Plant materials for extraction								
	Extraction of *Z. talanai* leaf extracts								
	Biosynthesis of Silica NPs infused with extracts								
	Characterization of Silica NPs infused with the extract								
	Data Processing/Analysis								
2	**Optimization Studies (Cytotoxicity and Molluscicidal Evaluation)**								
	Cytotoxicity Evaluation								
	Molluscicidal Evaluation Dipping Method Tray Experiment								
	Data Processing/Analysis								
3	**Pellet Formulation/Development**								
	Data Processing/Analysis								

ANNEX 2. WORKPLAN SCHEDULE *continued. . .*

Study	Activity	2023				2024			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
4	**Laboratory and Field Experiments: Cytotoxic and Molluscicidal Activity of Balakat-Nano molluscicide Pellets against Invasive Apple Snails**								
	Cytotoxicity Evaluation (Laboratory)								
	Molluscicidal Evaluation: Dipping Method Tray Method (Laboratory)								
	Molluscicial Experiment: Small-scale Field Trials (Phase I)								
	Molluscicidal Experiment: Large-scale field trials (Phase II)								
	Data Processing/Analysis								
5	**Ecotoxicological tests of the Nano-molluscicide on Non-target Organisms**								
	Ecotoxicological assessment of the derived products on non-target fauna such as fishes, frogs, crabs, and insect species								
6	**Histological Studies and Structural Alterations in Footpad and Mantle Treated with Nano-molluscicide Pellet**								
	Tissue Processing / Slide Preparation and Reading of Slides								
	Data Processing/Analysis								
	Field Day, Training and Seminars								
	Report Writing								
	Application of copyright registration for the product derived in the study								